Statistical Variables

Introduction: Data Is Everywhere

From the moment you wake up to the time you go to sleep, you encounter data in countless forms. The number of steps you walked, your favorite food, the temperature outside, your score on a quiz, the time it takes to travel to school — all of these are pieces of information called **data**. But not all data is the same. Some data are numbers, like your height or age. Other data are words or categories, like your hair color or the brand of your shoes.

Understanding data is essential because it helps us make better decisions. Should the school canteen offer more rice meals or sandwiches? Which measure best represents your class's performance on a test? Is the temperature in your city stable or does it change dramatically from day to day?

This reading material examines three important topics in statistics: **types of data and levels of measurement**, **measures of central tendency** (mean, median, mode), and **measures of variability** (range, interquartile range, variance, and standard deviation). These tools allow us to collect, organize, summarize, and interpret data effectively.

Part 1: Types of Data

In statistics, **data** refers to pieces of information collected for analysis. Understanding the type of data helps us choose the correct method for collecting, organizing, analyzing, and interpreting it.

Qualitative Data (Categorical Data)

Definition: Data that describe qualities or characteristics. They cannot be measured numerically but can be grouped into categories.

Examples:

- Hair color: black, brown, blonde
- Gender: male, female
- Type of school: public, private
- Favorite subject: Math, Science, English

Quantitative Data (Numerical Data)

Definition: Data that deal with numbers and amounts. They can be counted or measured and support mathematical operations.

Examples:

- Age: 16 years old
- Number of learners in class: 45
- Weight: 52.5 kg
- Income: ₱25,000 per month

Part 2: Levels of Measurement

Once we identify a variable as qualitative or quantitative, we also determine its **level of measurement**. There are four levels: nominal, ordinal, interval, and ratio, arranged from least to most informative.

Nominal Level (Categorical)

Definition: Data are labeled or categorized without any implied order or ranking.

Examples:

- Blood type: A, B, AB, O
- Type of fruit: apple, banana, mango
- Region: NCR, CAR, Region IV-A

Ordinal Level (Rank)

Definition: Data that can be ordered or ranked, but differences between ranks are not measurable.

Examples:

- Survey rating: satisfied, neutral, dissatisfied
- Class ranking: 1st place, 2nd place, 3rd place
- Socioeconomic status: low, middle, high

Interval Level

Definition: Numerical data with measurable and meaningful differences between values, but no true zero (zero does not mean absence).

Examples:

- Temperature in Celsius: 30°C, 20°C (0°C does not mean no temperature)
- Dates: 2000, 2010
- IQ scores: 90, 100, 110

Ratio Level

Definition: Similar to interval level, but with an absolute zero. Zero means "none" or "nothing."

Examples:

- Height: 160 cm (0 cm means no height)
- Weight: 60 kg (0 kg means no weight)
- Age: 18 years (0 years means no age)
- Income: ₱30,000 per month (0 means no income)

Summary Table: Levels of Measurement

Level	Type	Order	Measurable Differences	True Zero	Example
Nominal	Qualitative	No	No	No	Eye color
Ordinal	Qualitative	Yes	No	No	Class rank
Interval	Quantitative	Yes	Yes	No	Temperature (°C)
Ratio	Quantitative	Yes	Yes	Yes	Height, weight

Part 3: Measures of Central Tendency

Measures of central tendency are statistical tools used to describe the center or typical value of a set of data. They provide a single value that represents the entire distribution. The three most common measures are: **mean** (average), **median** (middle value), and **mode** (most frequent value).

A. Mean (Average)

Definition: The sum of all values divided by the number of values.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

When to use: Used when data is uniform and free from extreme values (outliers). Commonly used for grades, temperatures, and performance scores.

Strengths: Uses all data values; widely understood.

Weaknesses: Sensitive to outliers; extreme values can distort the result.

Example: Quiz scores: 9, 8, 7, 10, 9, 8, 6, 10

Mean = (9+8+7+10+9+8+6+10) ÷ 8 = 67 ÷ 8 = 8.375

Interpretation: The mean score is 8.375 out of 10, suggesting that, on average, learners performed well.

B. Median

Definition: The middle value of an ordered data set. If there is an even number of values, the median is the average of the two middle values.

When to use: Best used when data is skewed or has outliers, such as income distributions or test scores with very low or high extremes.

Strengths: Unaffected by outliers; reliable for skewed data.

Weaknesses: Does not use all values; may ignore useful information.

Example: Scores: 45, 48, 50, 50, 50, 52, 55, 97, 100

Ordered: 45, 48, 50, 50, 50, 52, 55, 97, 100 (9 values)

Median = 5th value = 50

Interpretation: Half of the learners scored below 50 and half scored above 50. The median is not affected by the two unusually high scores (97 and 100).

C. Mode

Definition: The value that appears most frequently in a data set.

When to use: Best used with categorical or nominal data (e.g., favorite color, shoe size) or when identifying the most common item is important.

Strengths: Only measure that can be used with qualitative data; easy to identify.

Weaknesses: A data set may have no mode, one mode, or multiple modes.

Example: Shirt sizes sold: M, L, M, XL, M, S, L, M

Frequency: S=1, M=4, L=2, XL=1

Mode = M (medium)

Interpretation: Medium-sized shirts were the most commonly sold, which is useful for inventory planning.

Choosing the Right Measure of Central Tendency

Situation	Best Measure	Reason
Data is balanced, no outliers	Mean	Uses all data; gives accurate average
Data has outliers (very high or low values)	Median	Not affected by extreme values
Data is categorical (words, not numbers)	Mode	Only measure that works with categories
Finding most common item	Mode	Shows what appears most often

Computing Mean, Median, and Mode Using Technology

Using MS Excel:

Measure	Formula
Mean	=AVERAGE(range)
Median	=MEDIAN(range)
Mode	=MODE.SNGL(range)

Using Jamovi:

1. Copy data into Jamovi
2. Click Analyses → Exploration → Descriptives
3. Move variable to Variables panel
4. Check Mean, Median, Mode boxes

Part 4: Measures of Variability

Measures of variability describe the spread or dispersion of data — how far apart data points are from each other and from the mean. While measures of central tendency tell us the typical value, measures of variability tell us how much the data fluctuates around that value.

A. Range (R)

Definition: The difference between the highest and lowest values in a dataset.

Formula:

$$R = \text{Highest Value} - \text{Lowest Value}$$

When to use: Helpful when understanding extreme values is important.

Strengths: Very easy to compute.

Weaknesses: Does not account for how values are distributed; affected by outliers.

Example 1: Test scores from 14 to 39 → R = 39 - 14 = 25

Example 2: Minutes to solve a problem: 23, 10, 18, 19, 43, 16 → R = 43 - 10 = 33

B. Interquartile Range (IQR)

Definition: The spread of the middle 50% of the data. It is the difference between the third quartile (Q3, 75th percentile) and the first quartile (Q1, 25th percentile).

Formula:

$$IQR = Q_3 - Q_1$$

When to use: Useful when we want to measure variability while minimizing the impact of extreme values (outliers).

Strengths: Resistant to outliers; focuses on the central portion of data.

Example: Test scores: 35, 32, 37, 45, 15, 23, 26, 26, 39, 40

Step 1: Arrange the data in ascending order.

15, 23, 26, 26, 32, 35, 37, 39, 40, 45

Step 2: Find Q1 (median of the lower half).

Lower half (first 5 values): 15, 23, 26, 26, 32
Q1 = 26

Step 3: Find Q3 (median of the upper half).

Upper half (last 5 values): 35, 37, 39, 40, 45
Q3 = 39

Step 4: Compute IQR.

IQR = Q3 - Q1 = 39 - 26 = 13

Interpretation: The middle 50% of learners' test scores are spread across 13 points. Scores are not extremely scattered.

C. Variance

Definition: The mean of squared differences from the average. It measures how spread out data is from the mean.

Formulas:

Sample variance:

$s^2 = \sum(X - \bar{X})^2 / n - 1$

Population variance:

$\sigma^2 = \sum(X - \mu)^2 / N$

When to use: Useful in advanced statistical analysis and comparing spread between datasets.

Example: Using the same test scores in previous exercise

Data: 35, 32, 37, 45, 15, 23, 26, 26, 39, 40

Step 1: Find the mean ($\bar{X}$).

Sum = 35 + 32 + 37 + 45 + 15 + 23 + 26 + 26 + 39 + 40 = 318

Number of values (n) = 10

$\bar{X} = 318/10 = 31.8$

Step 2 & 3: Compute deviation and squared deviation directly.

Score (X)	Deviation $(X-\bar{X})$	Squared Deviation $(X-\bar{X})^2$
35	35 - 31.8 = 3.2	3.2^2 = 10.24
32	32 - 31.8 = 0.2	0.2^2 = 0.04
37	37 - 31.8 = 5.2	5.2^2 = 27.04
45	45 - 31.8 = 13.2	13.2^2 = 174.24
15	15 - 31.8 = -16.8	$(-16.8)^2$ = 282.24
23	23 - 31.8 = -8.8	$(-8.8)^2$ = 77.44
26	26 - 31.8 = -5.8	$(-5.8)^2$ = 33.64
26	26 - 31.8 = -5.8	$(-5.8)^2$ = 33.64
39	39 - 31.8 = 7.2	7.2^2 = 51.84
40	40 - 31.8 = 8.2	8.2^2 = 67.24

Step 4: Sum the squared deviations.

$$\sum(X-\bar{X})^2=10.24+0.04+27.04+174.24+282.24+77.44+33.64+33.64+51.84+67.24=757.60$$

Step 5: Divide by $(n-1)$ for sample variance.

$s^2 = 757.60/(10-1) = 84.18$

Final Answer: Variance = **84.18**

Interpretation: The test scores have a moderate to high spread from the mean. Some learners performed much better or worse than others.

D. Standard Deviation

Definition: The square root of the variance. It indicates how scattered data values are around the mean, measured in the same units as the original data.

Formulas:

Sample standard deviation:

$s = \sqrt{s^2} = (\sum(X - \bar{X})^2 / n - 1)^{1/2}$

Population standard deviation:

$\sigma = \sqrt{\sigma^2} = (\sum(X - \mu)^2 / N)^{1/2}$

Interpretation:

- **High standard deviation** → Data points are spread out over a wide range from each other and from their own mean → greater variability
- **Low standard deviation** → Data points are clustered closely together and near their own mean → more consistency

Summary Table: Measures of Variability

Measure	What It Tells Us	Best Used When	Affected by Outliers?
Range	Spread between highest and lowest	Quick estimate of spread	Yes
IQR	Spread of middle 50%	Data has outliers	No
Variance	Average squared distance from mean	Comparing spread between datasets	Yes
Standard Deviation	Typical distance and spread of data from mean (original units)	Interpreting spread in real-world context	Yes

Computing Measures of Variability Using Technology

Using MS Excel:

Measure	Formula
Range	=MAX(range) - MIN(range)
IQR	=QUARTILE(range,3) - QUARTILE(range,1)
Variance	=VAR.S(range)
Standard Deviation	=STDEV.S(range)

Using Jamovi:

1. Copy data into Jamovi
2. Click Analyses → Exploration → Descriptives
3. Move variable to Variables panel
4. Check Range, IQR, Variance, Standard Deviation boxes

Real-Life Applications

Scenario	Appropriate Measure	Why
Comparing consistency of two machines	Standard deviation	Smaller SD = more consistent
Analyzing household income with billionaires	Median or IQR	Not affected by extreme outliers

Scenario	Appropriate Measure	Why
Checking temperature stability for crops	Range or standard deviation	Shows how much temperature fluctuates
Finding most popular shoe size	Mode	Identifies what customers buy most
Reporting average class test score (no outliers)	Mean	Uses all data, widely understood

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Key Terms

Term	Definition
Data	Pieces of information collected for analysis
Qualitative Data	Data that describes qualities or characteristics (words/categories)
Quantitative Data	Data that represents numerical values (numbers)
Nominal Level	Categories with no order (e.g., blood type, color)
Ordinal Level	Categories with order but no measurable difference (e.g., class rank)
Interval Level	Numerical data with meaningful differences but no true zero (e.g., temperature in °C)
Ratio Level	Numerical data with meaningful differences and a true zero (e.g., height, weight)

Term	Definition
Mean	The average; sum of values divided by number of values
Median	The middle value when data is ordered
Mode	The most frequently occurring value
Outlier	An extreme value that is very different from the rest of the data
Range	The difference between the highest and lowest values
Interquartile Range (IQR)	The spread of the middle 50% of data (Q3 - Q1)
Variance	The average of squared differences from the mean
Standard Deviation	The square root of variance; typical distance from the mean
Variability	The spread or dispersion of data points

Review Questions

Write your answers on a separate sheet of paper.

Types of Data and Levels of Measurement

1. Classify each as qualitative or quantitative: (a) favorite color, (b) height in cm, (c) number of siblings, (d) brand of shampoo.
2. Identify the level of measurement (nominal, ordinal, interval, ratio): (a) temperature in °C, (b) class ranking, (c) weight in kg, (d) blood type.
3. A survey asks: "Rate your satisfaction from 1 (very dissatisfied) to 5 (very satisfied)." What level of measurement is this?
4. Why can't you calculate the mean of hair colors?

Measures of Central Tendency

5. Find the mean, median, and mode of: 12, 15, 12, 18, 20, 12, 16.
6. A teacher recorded quiz scores: 50, 48, 49, 50, 45, 50, 42, 10, 50, 38. Which measure (mean, median, or mode) best represents the class performance? Why?
7. A shoe store wants to know which size to restock most often. Should they use mean, median, or mode? Why?
8. If a dataset has an outlier, why is the median preferred over the mean?

Measures of Variability

9. Find the range of: 23, 45, 67, 12, 89, 34, 56.
10. What does a small standard deviation tell you about a dataset?
11. Two sections have the same mean score on a test. Section A has a standard deviation of 5, while Section B has a standard deviation of 12. Which section has more consistent scores? Why?
12. Why is the interquartile range (IQR) better than the range when outliers are present?

Mixed Problems

13. A researcher collected the following data on weekly allowance (in PHP) of 10 students: 150, 200, 180, 500, 190, 210, 175, 185, 195, 5000. (a) Which measure of central tendency is most appropriate? (b) Compute it. (c) What is the range? (d) What does the range tell you?

14. A farmer recorded daily mango harvests (kg) for 7 days: 20, 22, 24, 21, 23, 25, 22. (a) Compute the mean. (b) Compute the standard deviation. (c) Interpret the standard deviation.
15. Explain the difference between variance and standard deviation. Why is standard deviation often easier to interpret?

Answer Key

Types of Data and Levels of Measurement

1. (a) qualitative, (b) quantitative, (c) quantitative, (d) qualitative
2. (a) interval, (b) ordinal, (c) ratio, (d) nominal
3. Ordinal (ordered categories but differences between ranks are not necessarily equal)
4. Hair colors are qualitative (categorical) data; mean requires numerical values.

Measures of Central Tendency

5. Mean = (12+15+12+18+20+12+16) ÷ 7 = 105 ÷ 7 = 15; Median = 15 (ordered: 12,12,12,15,16,18,20); Mode = 12
6. Median is best because the score of 10 is an outlier that pulls the mean down. Median = 48.5, Mean ≈ 43.2, Mode = 50
7. Mode — they want the most frequently purchased size.
8. The median is not affected by extreme values; the mean is pulled toward the outlier.

Measures of Variability

9. Range = 89 - 12 = 77
10. A small standard deviation means data points are clustered closely around the mean (low variability, high consistency).
11. Section A (SD = 5) has more consistent scores because the standard deviation is smaller.
12. IQR focuses on the middle 50% of data and ignores extreme values (outliers).

Mixed Problems

13. (a) Median is most appropriate because 5000 is an outlier. (b) Ordered: 150,175,180,185,190,195,200,210,500,5000; Median = (190+195)÷2 = 192.5 PHP (c) Range = 5000 - 150 = 4850 (d) The range is very large, indicating extreme spread due to the outlier.

14. (a) Mean = (20+22+24+21+23+25+22) ÷ 7 = 157 ÷ 7 ≈ 22.43 kg (b) Standard deviation ≈ 1.72 kg (c) The daily harvest varies by about 1.72 kg from the mean, indicating relatively consistent production.
15. Variance is the average squared distance from the mean (units squared). Standard deviation is the square root of variance, measured in the same units as the original data, making it easier to interpret in real-world contexts.